Fiscal Year:	FY 2017	**Task Last Updated:**	FY 04/25/2018
PI Name:	Cromwell, Ronita Ph.D.		
Project Title:	Bed Rest Standard Measures		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058	**Congressional District:**	22
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	12/01/2004	**End Date:**	02/28/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Norsk, Peter	**Contact Phone:**	
Contact Email:	Peter.norsk@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 2/28/2017 per K. George/HRP (Ed., 11/30/16) NOTE: End date changed to 10/31/2016 per K. George/HRP (Ed., 8/16/16) NOTE: Changed end date to 7/31/2016 as work is continuing per PI (Ed., 12/10/15) NOTE: Change in end date per PI (Ed., 3/12/2015)		
Key Personnel Changes/Previous PI:	September 2012: David Ware was removed from the protocol. October 2010: Buccello-Stout, Regina (Universities Space Research Association) removed from protocol. Zanello, Susana (Universities Space Research Association) Added to protocol.		
COI Name (Institution):	Yarbough, Patrice Ph.D. (Universities Space Research Association) Zanello, Susana Ph.D. (Universities Space Research Association)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The Flight Analogs Project (FAP) Plan outlines a multi-phase approach for implementation of NASA-funded bed rest studies. This study is currently conducted at the NASA Flight Analogs Research Unit located at The University of Texas Medical Branch (UTMB) in Galveston, Texas. Although head-down bed rest has been used as a spaceflight analog to study changes in physiologic function associated with spaceflight for many years at many different institutions, standards for conducting such studies have not previously been established. Conditions of bed rest have been investigation or site specific. Most studies have had no direct comparisons of bed rest data with spaceflight data. As a consequence, systematic, multi-disciplinary comparisons across studies and comparisons of bed rest and actual spaceflight data have not been possible. To address these issues, the FAP has developed standard conditions for implementation of all NASA-funded bed rest studies and a set of standard measurements to be collected from all bed rest subjects. The standard measurements replicate, for the most part, measurements obtained on astronauts before and after long duration missions. Spaceflight and bed rest measurements are taken by the Johnson Space Center Laboratories and are used for three purposes: they make possible a direct comparison between responses to spaceflight and responses to head-down bed rest; they provide a characterization of human responses to long duration head-down bed rest; and they provide the ability to assess candidate countermeasures in a multi-disciplinary manner by evaluating the effects on the targeted systems as well as possible unexpected effects on non-targeted systems. The following measures are included in the standard measures complement. Nutritional status and immune function are assessed through blood, urine, and saliva samples. Standard clinical laboratory assessment of blood and urine are used to monitor subjects' health status throughout the study. Dual Energy X-Ray Absorptiometry (DXA) in combination with bone biomarkers are used to assess changes in bone. Physical Fitness is assessed through measures of muscle strength, endurance, flexibility, and maximum aerobic capacity. Posturography and monosynaptic T-reflex testing provides assessments of neurological function. The cardiovascular system is assessed through measures of orthostatic tolerance, blood volume, and cardiac function. These multidisciplinary measures are provided to investigators to enhance interpretation of system-specific results. By standardizing conditions and measures for all bed rest studies, NASA is able to compare results across bed rest studies in a multidisciplinary manner. The standard set of measures will be collected on as many bed rest subjects as is feasible to provide a multi-discipline evaluation of potential countermeasures. These measures will be integrated with the peer-reviewed science requirements on a non-interference basis.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The Campaign 11 (C11) (Countermeasures and Functional Testing in Bed Rest) bed rest study was comprised of 8 individual investigations that participated in this 70-day, 6° head-down tilt bed rest campaign. These investigations were integrated for efficiency, and to allow investigators to share like measures. In addition to the investigator studies, standard measures were also collected. Standard measures targeted a wide range of physiologic systems and represent some of the testing routinely done on astronauts. Standard measures allowed for comparisons across studies as they are collected on all subjects who participated in the bed rest campaign. Bed rest subjects underwent rigorous medical and psychological screening. Standardized conditions included 70 days of bed rest in a 6° head-down tilt position. Subjects' vital signs, body weight, and fluid intake and output were measured daily. A standardized diet was provided to ensure consistent nutritional intake across subjects. Exercising subjects were prescribed individualized aerobic and resistance training 6 days per week performed in a horizontal body position. Subjects in the testosterone supplementation countermeasure group received testosterone enanthate injections at 2-week intervals during bed rest. A total of 37 subjects participated in the C11 bed rest campaign. Of these subjects, 11 were controls (CON) and did not complete an exercise protocol. Ten subjects completed the integrated resistance and aerobic training exercise (Ex) using equipment that permitted exercise in a horizontal position. Eight subjects completed exercise with testosterone supplementation (Ex+T) and 8 subjects completed the exercise protocol using a flywheel device. One female subject participated prior to the addition of the Ex+T study. Once the Ex+T study was added to the complement, the remaining subjects were all males so that comparisons to the Ex+T subjects were possible. Results of some of the standard measures are provided here. For knee extension isokinetic strength, CON subjects showed a significant decrease in strength as compared to EX subjects when tested 2 days and 12 days post-bed rest (BR+2 and BR+12) (BR+2: CON 156+36 Nm; EX 193+25 Nm) (BR+12: CON 169+32 Nm; Ex 201+24 Nm). The EX subjects demonstrated significantly greater vertical jump height than CON on the day subjects stood up from bed (BR+0) (CON 32.0+5.8 cm; Ex 41.1+5.5 cm). Both EX and EX+T subjects showed greater vertical jump power than CON subjects on BR+0 (CON 36.5+4.0 W/Kg; Ex 43.8+6.7 W/Kg; Ex+T 44.1+8 W/Kg). For aerobic capacity (VO2peak), EX and EX+T subjects showed greater VO2peak than CON subjects on BR+0 (CON 2.2+0.4 L/min; Ex 3.2+0.6 L/min; Ex+T 3.1+0.4 L/min). Plasma volume measures demonstrated significant decreases in all groups by bed rest day 7 (BR7) as compared to pre-bed rest values (BR-2) (BR-2: CON 3.0+0.4 L; EX 3.2+0.5 L; Ex+T 3.0+0.4 L) (BR7: CON 2.5+0.4 L; Ex 2.7+0.5 L; Ex+T 2.7+0.4 L). On BR+0, plasma volume showed significant decrease from pre-bed rest for CON and Ex subjects (BR+0: CON 2.7+0.5 L; Ex 2.8+0.4 L). Further analyses of these measures can be found in publications resulting from this bed rest campaign. Analyses continue and further publications are being prepared.
Bibliography Type:	Description: (Last Updated: 05/16/2019)
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